



Internship Reference

Mr Max Jungmann, born on 28 March 2002, was employed in our company from 1 March 2025 to 31 May 2025 as an Intern.

Lufthansa Systems GmbH is one of the world's leading IT providers in the airline industry. With a combination of in-depth airline know-how, many years of project experience, and technological expertise, the company offers a comprehensive range of successful, often market-leading products for the aviation industry that more than 350 airline customers' trust. The portfolio includes innovative IT products and consulting services that offer high economic benefits and help to increase efficiency. In addition to its headquarters in Raunheim near Frankfurt/Main, Lufthansa Systems is represented in 16 countries.

Mr Jungmann completed an internship in the department Product Development Scheduling in Raunheim. During his three-month internship, Mr Jungmann supported our algorithm team in the development of a new stochastic optimization approach for the Rotation Optimizer. The Rotation Optimizer is a software solution that optimizes flight schedules with respect to several user-defined objectives. Its purpose is to route aircraft under consideration of multiple KPIs, including on-time performance.

During his internship, Mr Jungmann was given the task of exploring and implementing an alternative, stochastic optimization approach to increase on-time performance and to compare its solution quality with the legacy approach that was already in use. He developed two versions of the new approach for the problem: a basic version and a more sophisticated version that built upon the basic version. His work consisted mainly of mathematical modeling (especially integer programming) as well as implementing mathematical ideas in Python, using libraries like pandas, polars and google-ortools. To organize our work the optimizer team uses classical software engineering tools like GitHub, Continuous Integration and Code Quality Checks. It turned out that even the basic version of the new approach led to a significant improvement over the legacy approach. It reduced the total arrival delay by a factor of two compared to the legacy approach. Moreover, the more sophisticated approach could again slightly improve on the basic approach but with the cost of a much higher running time. Finally, Mr Jungmann also presented the results to a broader audience, including other colleagues and customers.

We decided to first incorporate the basic on-time performance optimization model into our software product. At the end of the internship, the Rotation Optimizer with the new on-time performance functionality was successfully tested on live data of one of our biggest customers. Therefore, we have made the new functionality developed by Mr Jungmann available to all our customers.

In each stage of the internship, Mr Jungmann consistently acquired excellent specialist knowledge in an impressively short time and was always able to incorporate and implement it very successfully in his daily work.



Mr Jungmann already came equipped with a deep and reliable understanding of combinatorial optimization. In particular, he was already familiar with the theory of stochastic optimization and had some practice in the use of solver libraries in the context of python programming. He grasps even high level abstract mathematical ideas almost instantly and explores them with ease. During his internship he acquired many additional skills in advanced usage of data science libraries (polars, pandas), data visualization and the development of mathematical software as a part of a team, using version control and continuous integration. He picked up all those skills at an incredibly fast pace. At the same time, he was able to present his accomplishments confidently and convincingly to a large group of clients and colleagues. Mr Jungmann is an all-around pleasure to work with.

He had an excellent ability to familiarise himself rapidly, which always enabled him to grasp even the most complex work content very well within a very short time. Mr Jungmann always worked using his own initiative and completely identified with his responsibilities and our company at all times. He always displayed an impressive level of dedication and motivation. During the entire internship, Mr Jungmann stood out because of his extremely high and excellent eagerness to learn. He consistently completed all his tasks commendably, even under the most difficult working conditions and under extreme stress.

Mr Jungmann was always highly appreciated by our company because of his extremely diligent and responsible way of working. His work was always characterised by an extraordinary level of reliability.

He always found excellent solutions for any problems which arose. Mr Jungmann successfully implemented a superior approach to the optimization of on-time-performance for the rotation optimizer. His contribution is now part of our product and has a significant positive impact on the quality of solutions available to our customers. He gave a clear, well organized, and convincing presentation of his accomplishments to a group of customers and colleagues. Mr Jungmann always carried out the tasks assigned to him during the internship to our utmost satisfaction.

He was an extremely loyal employee who was very well integrated into the team. His behaviour towards his line managers, instructors, colleagues and customers was always exemplary.

The collaboration with Mr Jungmann was very pleasant and gratifying. We believe he is very well-suited for his chosen profession. We would like to thank him for his consistently very good performance and wish him continued success and all the best both for his future career and private life.

Raunheim, 31 May 2025
Lufthansa Systems GmbH



J. V. Jan-Peter Gänse
Head of NetLine